Cheatsheet AZ-305 — 16 · Additional compute, Azure Arc and Infrastructure as Code

1. Compute services that were missing

Service	What it is	Signal
Azure Virtual Desktop (AVD)	Managed multi-session Windows 10/11 remote desktops and apps; host pools (pooled / personal), application groups (Desktop / RemoteApp), workspaces; **FSLogix** for profiles on Azure Files/NetApp	"remote desktops", "multi-session", "profiles", "publish RemoteApps"
Azure Batch	Large-scale parallel compute with its **own scheduler**; VM pools (incl. Spot), jobs and tasks	"parallel processing", "render", "no third-party scheduler"
Azure CycleCloud	Orchestration of HPC clusters with **third-party schedulers** (Slurm, PBS, LSF)	"HPC", "third-party scheduler"
HPC (H/HB/HC series, InfiniBand)	VMs for MPI / simulation	"low-latency interconnect", "MPI"
Azure Spring Apps	Managed PaaS for **Java Spring Boot** (Enterprise: Tanzu)	"Spring Boot microservices without managing infrastructure"
Static Web Apps	Hosting for static frontends (React, Angular…) with **integrated Functions** as the API, CI/CD from GitHub/Azure DevOps, global CDN, built-in auth	"static site + serverless API", "JAMstack", "deploy from GitHub"
App Service (Linux / containers)	Web apps from a container image	"custom container in App Service"
Azure Functions Durable / Logic Apps	See cheatsheet 06	
Dedicated Hosts	Dedicated physical servers	"no shared hardware"
Azure Stack HCI / Hub / Edge	Azure in your datacenter (HCI: hyperconvergence; Hub: Azure services on-prem; Edge: AI/IoT appliance)	"run Azure services on-premises", "disconnected"
Azure Local	Current name of Azure Stack HCI	

2. Azure Arc

Capability	What it provides
Arc-enabled servers	Windows/Linux servers **on-prem or in other clouds** as Azure resources: RBAC, tags, Policy (guest configuration), **Azure Monitor Agent**, Defender for Servers, Update Manager, Automation, extensions
Arc-enabled Kubernetes	External clusters managed from Azure: GitOps (Flux), Policy, Defender, Monitor
Arc-enabled data services	**Azure SQL Managed Instance** and PostgreSQL **running on your on-prem Kubernetes**, with Azure billing and management; direct or indirect (disconnected) mode
Arc-enabled SQL Server	On-prem SQL Server inventoried and protected (Defender, best practices, Purview)
Arc-enabled VMware / SCVMM	Manage VMware/Hyper-V VMs from Azure
Signal	"on-premises servers", "AWS instances", "single pane", "apply Azure Policy to on-prem", "monitor hybrid with AMA"

- Arc **does not migrate** and **does not replicate**: it manages resources where they are.
- Arc + AMA + central Log Analytics = hybrid monitoring (cheatsheet 07).
- Arc-enabled servers require the **Connected Machine agent** and outbound HTTPS to Azure.

3. Infrastructure as Code

Tool	What it is	Detail
ARM templates	Native declarative JSON	Verbose; the foundation of everything
Bicep	Microsoft's declarative DSL that compiles to ARM	More readable, modules, types; **recommended**; stateless (Azure is the source of truth)
Terraform	Multi-cloud (HashiCorp), HCL, **state file**	Multi-cloud or the team already uses it; state in Storage with locking
Pulumi	IaC with programming languages	Niche
Template Specs	ARM/Bicep templates **stored and versioned as Azure resources**, shareable via RBAC	Replaces Blueprints for template reuse
Deployment Stacks	Group deployed resources as a managed unit; **deny settings** protect against changes outside the stack; cascading delete	Replaces Blueprints for lifecycle governance
Azure Blueprints	Retired (July 2026); definition at MG, assignment per subscription	See cheatsheet 04 if it appears
Deployment modes	**Incremental** (default: adds/updates, does not delete) vs **Complete** (deletes what is not in the template)	"remove resources not in template" → Complete
Deployment scopes	Resource group · Subscription · Management group · Tenant	Policy/RBAC at MG level → deployment at MG scope
What-if	Preview changes	"preview changes before deploying"

Tool	What it is	Detail
Azure Resource Graph	KQL queries over all resources	"query resources across subscriptions"

- **Configuration version history** → templates (Bicep/ARM) in **Git** (Azure Repos / GitHub).
- Configuration **inside** the VM at provisioning → **Custom Script / DSC / Azure Automation State Configuration**, extensionProfile in VMSS; **Azure Image Builder / Compute Gallery** for golden images.
- **Azure App Configuration** + Key Vault references → centralized app configuration and feature flags.

4. DevOps and deployment

Piece	Detail
Azure DevOps	Repos (Git), Pipelines (YAML), Boards, Artifacts, Test Plans
GitHub	Actions (CI/CD), Advanced Security, Codespaces
Service connection / workload identity federation	Pipelines authenticate to Azure **without secrets** (OIDC)
Environments + approvals	Gates between stages
Deployment slots (App Service)	Swap with no downtime; warm-up; rollback
Blue/green and canary	Slots, Traffic Manager weighted, Front Door, Container Apps revisions, Istio
Rolling upgrade (VMSS)	Update in batches with health probe
Feature flags	App Configuration
Self-hosted agents	When the pipeline needs a private network / specific software
Artifacts / Container Registry	Versioned packages and images; ACR **Tasks** for builds; geo-replication (Premium)

5. VM images and configuration

Need	Solution
Reusable, versioned, multi-region golden image	**Azure Compute Gallery** (generalized/specialized images, replicas per region)
Automate image creation	**Azure Image Builder**
Post-boot configuration	Custom Script Extension, DSC, cloud-init (Linux)
Continuous desired state	Azure Automation State Configuration / **Machine Configuration (Policy guest configuration)**
Patching	**Azure Update Manager** (Azure + Arc)
Inventory and changes	Change Tracking and Inventory

6. Decision tree

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What is asked for?
├─ Multi-session remote desktops / apps ..... Azure Virtual Desktop
├─ Managed parallel compute ................. Batch
├─ HPC with Slurm/PBS ....................... CycleCloud
├─ Spring Boot without managing infra ....... Azure Spring Apps
├─ Static frontend + serverless API from GitHub ..... Static Web Apps
├─ Manage servers/K8s/SQL outside Azure ..... Azure Arc
├─ SQL MI on my on-prem Kubernetes .......... Arc-enabled data services
├─ Define infrastructure with history ....... Bicep in Git
├─ Multi-cloud IaC .......................... Terraform
├─ Reuse templates with RBAC ................ Template Specs
├─ Protect resources from changes outside the deployment .... Deployment Stacks (deny settings)
├─ Delete whatever is not in the template ... Complete mode
├─ Pipeline without secrets ................. Workload identity federation
├─ Zero-downtime deployment on App Service .. Slots
└─ VM image versioned per region ............ Compute Gallery
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7. Signals → answer

Wording	Answer
"multi-session Windows desktops", "FSLogix profiles"	Azure Virtual Desktop
"render farm, thousands of tasks, no scheduler"	Batch
"Spring Boot", "Tanzu"	Azure Spring Apps
"React app + Azure Functions API, GitHub"	Static Web Apps
"apply Azure Policy to AWS EC2 instances"	Azure Arc-enabled servers
"SQL Managed Instance on-premises Kubernetes"	Arc-enabled data services
"readable, modular, Azure-only IaC"	Bicep
"state file", "multi-cloud"	Terraform
"share templates with teams via RBAC"	Template Specs
"prevent deletion outside the deployment"	Deployment Stacks

Wording	Answer
"delete resources not in template"	Complete mode
"preview changes"	What-if
"query all VMs across 50 subscriptions"	Azure Resource Graph
"golden image, multiple regions"	Compute Gallery
"patch Azure and on-prem servers centrally"	Update Manager (+ Arc)

8. Mistakes to avoid

1. Batch for HPC with a third-party scheduler → CycleCloud.
2. Arc as a migration tool → it only manages.
3. Blueprints as the modern answer for reuse → Template Specs / Deployment Stacks.
4. Incremental when asked to delete what is left over → Complete.
5. ZIP or images as "configuration definition with history" → Bicep in Git.
6. Compute Gallery for App Service → it is for VMs.